

Thermal Forcing and Coral Bleaching Resilience in the Ryukyu Archipelago: A Multi-Site Survey, 2020–2024

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Abstract. Coral bleaching events have increased in frequency and severity across the Indo-Pacific, yet resilience patterns remain undersampled at mid-latitude reef systems. Between 2020 and 2024 we conducted quarterly surveys at 12 reef sites spanning 24°N–27°N in the Ryukyu Archipelago, combining in-situ sea-surface temperature (SST) loggers, quadrat-based bleaching indices, and symbiont community profiling. Bleaching index scales nonlinearly with cumulative thermal anomaly (degree heating weeks, DHW), with a critical threshold near $DHW = 8.4$ above which mean bleaching exceeds 60%. Two northern sites (Cape Hedo, Sesoko West) maintained <20% bleaching during the 2023 Pacific marine heatwave despite $DHW > 10$, suggesting local thermal adaptation or symbiont community buffering. These refugia are priority candidates for Pacific-wide conservation corridor planning under RCP 4.5 and RCP 8.5 trajectories. We discuss implications for managed relocation of resilient genotypes and identify three unresolved questions concerning the role of background microbial community structure in buffering acute thermal stress.

Keywords: coral bleaching, Ryukyu Archipelago, thermal resilience, degree heating weeks, symbiont shuffling, climate refugia

1. Introduction

The global coral crisis has accelerated through the past three decades, with three mass bleaching events on the Great Barrier Reef alone since 2016 (Hughes et al. 2017; Dietzel et al. 2020). While tropical reef systems have received the bulk of monitoring attention, mid-latitude reefs in the West Pacific—long hypothesised to serve as climate refugia—remain sparsely sampled at multi-year resolution.

The Ryukyu Archipelago extends roughly 1,000 km from Kyushu to Taiwan, straddling the Kuroshio Current. Its reefs experience strong seasonal SST amplitude (14–30 °C) and episodic heat anomalies driven by the Western Pacific Warm Pool teleconnection. Despite their biogeographic importance, no standardised multi-site bleaching time series at quarterly cadence has been published for the archipelago as a whole.

Here we report on four years (2020 Q1 – 2024 Q4, 16 quarters) of paired physical and biological observations across 12 sites, with three questions: (i) what is the shape of the bleaching–thermal forcing relationship; (ii) are there sites that consistently depart from the regional norm; and (iii) do such resilience outliers share measurable predictors (depth, exposure, symbiont community)?

2. Methods

2.1 Study sites. Twelve sites were selected across four island groups (Yonaguni, Ishigaki, Okinawa-honto, Amami-oshima) spanning 24.3°N–27.0°N (Table S1). Site selection balanced forereef vs. backreef, exposure (NE, SW, leeward), and depth (5–18 m). Full coordinates and survey design are listed in the supplementary `dive-sites.csv`.

2.2 Temperature logging. Onset HOBO U22-001 loggers were deployed at each site at 8 m depth, recording at 30-minute intervals. Loggers were swapped and calibrated annually against an NIST-traceable reference. Degree heating weeks (DHW) were computed following NOAA Coral Reef Watch v3.1, using the site-specific climatology from 2010–2019.

2.3 Bleaching index. At each site, five permanent 25 m transects were photographed quarterly using a Canon R5 in a Nauticam housing with dual Inon Z-330 strobes. Quadrats (1 m², n = 50 per transect) were scored in Adobe Lightroom against the standardised Chasing Coral colour reference card. Bleaching index was defined as the percentage of live coral colonies exhibiting ≥2-grade colour loss (1–6 scale).

2.4 Symbiont community. At a subset of six sites, tissue biopsies (n = 12 per site per quarter) were preserved in 95% EtOH and shipped to UC Santa Cruz for ITS2 metabarcoding (primers `sym_VAR_5.8S2/sym_VAR_REV`). Operational taxonomic units were clustered at 97% identity against the GeoSymbio database.

2.5 Statistical analysis. We fit generalised additive mixed models (GAMMs) with site as a random effect and DHW, exposure, and depth as fixed effects, using a beta regression link appropriate for proportional data. All analyses were run in R 4.3 with the `mgcv` and `glmmTMB` packages; code is archived at github.com/wadachi/ryukyu-resilience.

3. Results

3.1 Thermal forcing. SST across the 12 sites ranged from 14.2 °C (Cape Hedo, Feb 2021) to 31.8 °C (Yonaguni, Aug 2023). Cumulative DHW exceeded the NOAA Alert Level 2 threshold ($DHW \geq 8$) at 9 of 12 sites during the 2023 heatwave, peaking at $DHW = 12.4$ at Ishigaki South.

3.2 Bleaching–DHW relationship. Mean bleaching index scaled nonlinearly with DHW (Fig. 2). A piecewise regression identified a breakpoint at $DHW = 8.36$ (95% CI 7.91–8.82), above which bleaching increased at 4.1× the baseline rate. Above $DHW = 10$, mean bleaching exceeded 60% across all sites pooled.

3.3 Resilience outliers. Two northern forereef sites—Cape Hedo and Sesoko West—maintained mean bleaching < 20% throughout 2023 despite $DHW > 10$ at both. Mixed-effects models attribute 38% of this residual variance to symbiont community evenness (Shannon H' , $p < 0.001$) and a further 17% to mean colony depth ($p = 0.004$).

3.4 Symbiont turnover. Six sites showed the expected *Acropora*–*Durusdinium* shuffle during peak DHW. The two resilience outliers instead hosted *Cladocopium* C3k dominance persistently, with no detectable shuffle. This divergence is the strongest candidate mechanism for the observed resilience.

4. Discussion

Our four-year time series confirms that bleaching in the Ryukyu Archipelago scales nonlinearly with cumulative thermal anomaly, consistent with regional patterns from the Great Barrier Reef and the Maldives. The breakpoint at $DHW \approx 8.4$ is statistically indistinguishable from the NOAA Alert Level 2 threshold, lending operational support to existing watch products.

The two resilience outliers are, to our knowledge, the first documented cases of sustained coral resistance to $DHW > 10$ in the Ryukyu Archipelago. Their shared signature—persistent *Cladocopium* C3k dominance, deeper mean colony depth, NE exposure—is consistent with the emerging 'background symbiont community' hypothesis (Boulotte et al. 2016; Cunning et al. 2018), which posits that thermal tolerance is set by standing symbiont diversity rather than post-stress shuffling.

Three caveats apply. First, our survey does not resolve post-bleaching mortality; resilient sites may still suffer sublethal fitness costs. Second, symbiont community is itself a moving target under RCP 8.5 warming trajectories. Third, the geographic breadth of resilience refugia may shrink faster than models predict if heatwaves outpace the rate of symbiont community turnover.

We close with a call for coordinated trans-pacific monitoring of symbiont community structure, alongside the existing temperature watch network. Without it, the conservation corridors we propose in Section 5 rest on inference from twelve sites—an unstable foundation for Pacific-wide planning.

5. Conclusion

Coral reefs of the Ryukyu Archipelago display a robust nonlinear response to thermal forcing, with two sites exhibiting resilient outliers consistent with the background-symbiont-community hypothesis. These refugia merit immediate conservation attention and long-term monitoring under projected climate trajectories.

Table 1. Site-level summary (2020–2024 mean).

Site	Mean SST (°C)	Mean DHW	Bleaching (%)	Dominant symbiont	Refugia?
Yonaguni West	27.4	9.1	62.3	Durusdinium	no
Ishigaki South	28.1	11.4	78.1	Durusdinium	no
Kabira North	27.9	9.8	55.6	Cladocopium C3	no
Sekisei Lagoon	28.4	10.9	71.4	Durusdinium	no
Cape Hedo	26.3	10.2	18.1	Cladocopium C3k	yes
Sesoko West	26.5	10.5	17.4	Cladocopium C3k	yes
Maeda Point	26.8	9.0	44.7	Cladocopium C3	no
Onna Village East	26.9	8.7	39.2	Cladocopium C3	no
Kerama Inner	27.1	8.4	36.8	Cladocopium C3	no
Kerama Outer	27.0	9.2	48.5	Cladocopium C3	no
Amami North	26.6	8.1	31.9	Cladocopium C3	no
Amami South	26.8	8.6	38.5	Cladocopium C3	no

Sites highlighted in amber are the two resilience refugia discussed in §3.3.

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